

SIMATS ENGINEERING

CO4-AT2

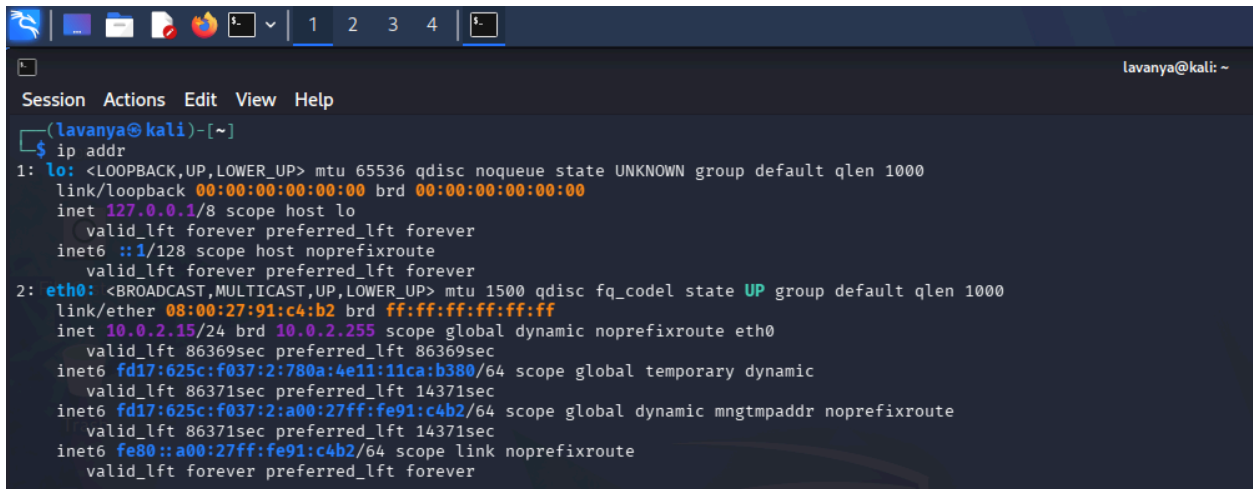
COURSE NAME: Ethical Hacking COURSE CODE: ITA1407

1. TJX – Wireless Security

COMMANDS

ip addr
iwconfig
nmcli device status
sudo ss -tuln
sudo iptables -L

OUTPUT



```
Session Actions Edit View Help
(lavanya@kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:91:c4:b2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86369sec preferred_lft 86369sec
    inet6 fd17:625c:f037:2:780a:4e11:11ca:b380/64 scope global temporary dynamic
        valid_lft 86371sec preferred_lft 14371sec
    inet6 fd17:625c:f037:2:a00:27ff:fe91:c4b2/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86371sec preferred_lft 14371sec
    inet6 fe80::a00:27ff:fe91:c4b2/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

```

(lavanya@kali)-[~]
$ iwconfig
Command 'iwconfig' not found, but can be installed with:
sudo apt install wireless-tools
Do you want to install it? (N/y)y
sudo apt install wireless-tools
[sudo] password for lavanya:
Installing:
  wireless-tools

Installing dependencies:
  libiw30t64

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 1066
  Download size: 102 kB
  Space needed: 294 kB / 29.5 GB available

Continue? [Y/n] y
Get:1 http://http.kali.org/kali kali-rolling/main amd64 libiw30t64 amd64 30~pre9-22 [18.0 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 wireless-tools amd64 30~pre9-22 [83.7 kB]
Fetched 102 kB in 12s (8,692 B/s)
Selecting previously unselected package libiw30t64:amd64.
(Reading database... 427927 files and directories currently installed.)
Preparing to unpack .../libiw30t64_30~pre9-22_amd64.deb...
Unpacking libiw30t64:amd64 (30~pre9-22)...
Selecting previously unselected package wireless-tools.
Preparing to unpack .../wireless-tools_30~pre9-22_amd64.deb...
Unpacking wireless-tools (30~pre9-22)...
Setting up libiw30t64:amd64 (30~pre9-22)...
Setting up wireless-tools (30~pre9-22)...
Processing triggers for kali-menu (2026.2.6)...
Processing triggers for libc-bin (2.42-16)...
Processing triggers for man-db (2.13.1-1)...

```

```

(lavanya@kali)-[~]
$ nmcli device status
DEVICE  TYPE      STATE      CONNECTION
eth0    ethernet  connected  Wired connection 1
lo      loopback  connected (externally)  lo

(lavanya@kali)-[~]
$ sudo ss -tln
Netid      State      Recv-Q      Send-Q      Local Address:Port      Peer Address:Port

```

```

(lavanya@kali)-[~]
$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination

```

1. Introduction

TJX Companies experienced a major payment-card data breach between 2005 and 2007. Attackers gained unauthorized access through weaknesses in the company's wireless network. The use of outdated WEP encryption, inadequate wireless access

controls, weak network segmentation, and insufficient monitoring contributed to the incident.

2. Vulnerabilities Identified

Vulnerability	Description	Risk
WEP encryption	WEP is outdated and can be cracked relatively easily	High
Weak wireless authentication	Unauthorized devices could gain network access	High
Poor network segmentation	Wireless users could reach sensitive internal systems	Critical
Insufficient monitoring	Suspicious wireless activity was not detected quickly	High
Payment-data exposure	Cardholder information was insufficiently protected	Critical

3. Impact

The attackers were able to access systems containing sensitive payment-card information. This resulted in financial losses, legal consequences, reputational damage, and risks to customers.

4. Recommendations

- Replace WEP with **WPA2-AES or WPA3**.
- Use **WPA2/WPA3-Enterprise with 802.1X authentication**.
- Implement strong individual credentials and, where appropriate, certificates.
- Separate wireless networks from payment-card systems using VLANs and firewalls.
- Deploy **Wireless Intrusion Detection/Prevention Systems (WIDS/WIPS)**.
- Detect unauthorized or rogue access points.
- Encrypt sensitive payment information both in transit and at rest.
- Apply relevant **PCI DSS** requirements.

- Continuously monitor logs and investigate suspicious activity.
- Regularly perform vulnerability assessments and penetration tests.

5. Conclusion

The TJX incident demonstrates that obsolete wireless encryption and poor network segmentation can expose highly sensitive information. Strong WPA2/WPA3 encryption, enterprise authentication, network isolation, wireless monitoring, and PCI DSS controls could have significantly reduced the risk and impact of the breach.